INCH-POUND

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MILITARY SPECIFICATION

SHOP EQUIPMENT, ORGANIZATIONAL REPAIR TRUCK-MOUNTED

This specification is approved for use by the Department of Army and is available for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a truck-mounted, organizational repair shop.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation .

SPECIFICATIONS

FEDERAL

O-A-548	- Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated
W-C-375	- Circuit Breakers, Molded Case, Branch Circuit and Service
TT-C-490	- Cleaning Methods For Ferrous Surfaces and Pretreatments For Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4940
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TT-P-664 - Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free
VV-B-680 - Brake Fluid, Automotive
VV-F-800 - Fuel Oil, Diesel
PPP-B-566 - Boxes, Folding, Paperboard
PPP-B-636 - Boxes, Shipping, Fiberboard
PPP-B-1055 - Barrier Material, Waterproofed, Flexible
PPP-P-40 - Preservation and Packing of Hand Tools; Tools and Tool Accessories For Power Driven, Metal and Woodworking Machinery
PPP-T-60 - Tape: Packaging, Waterproof
PPP-T-97 - Tape, Packaging/Industrial Filament Reinforced

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MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 - Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electronic, and Associated Equipment
MIL-V-173 - Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment)
MIL-P-514 - Plates, Identification, Instruction and Marking, Blank
MIL-T-704 - Treatment and Painting of Materiel
MIL-L-2104 - Lubricating Oil, Internal Combustion Engine, Tactical Service
MIL-L-2105 - Lubricating Oil, Gear, Multipurpose
MIL-C-3600 - Compressors, Rotary, Power-Driven; and Compressors, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and refrigerant), Packaging of
MIL-M-3971 - Meters, Time Totalizing, Non-Hermetically Sealed, Electrical: General Specification for
MIL-C-5541 - Chemical Conversion Coatings on Aluminum Alloys
MIL-M-7866 - Molybdenum Disulfide, Technical Lubrication Grade
MIL-E-10062 - Engines: Preparation for Shipment and Storage of
MIL-G-10924 - Grease, Automotive and Artillery
MIL-H-11511 - Heaters, Space: Multifuel, With Blower, 60,000 BTU/HR
MIL-A-11755 - Antifreeze, Arctic-Type
MIL-C-12000 - Cable, Cord, and Wire, Electric; Packaging of
MIL-C-13874 - Compressors, Reciprocating and Rotary, Air, Electric Motor Driven
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Support Parts: Packaging of
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of
MIL-M-18058 - Machinery, Metal and Woodworking Packaging of
MIL-L-23249 - Lathe, Engine and Tool Room, 13 through 17 inch
MIL-D-23377 - Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric
MIL-T-45542 - Tool Sets, Shop Sets and Kits Maintenance, Modification and Tool Packaging of
MIL-W-45562 - Welding and Soldering Equipment, Supplies, and Accessories, Packaging of

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MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic
MIL-C-53072 - Chemical Agent Resistant Coating (CARC) System Application Procedures and Quality Control Inspection
MIL-V-62038 - Vehicles, Wheeled: Preparation for Shipment and Storage of,

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-595 - Colors

MILITARY

MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-130 - Identification Marking of US Military Property
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-889 - Dissimilar Metals
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-1474 - Noise Limits for Army Materiel
MS21313 - Padlock Sets, Individually Keyed and Keyed Alike

(Unless otherwise indicated, copies of federal and military specifications are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those cited in the solicitation.

DRAWINGS

ARMY

TA13217E1000 - Shop Equipment, Organizational Repair, Truck-Mounted
13226E7428 - Usage Standardized Camouflage Pattern Painting for Shop Equipment, Organizational Repair Truck-Mounted
11023713 - Chassis Truck
11021065 - Welding Requirements
11021066 - Riveting Requirements

LIST OF COMPONENTS

ARMY

SC 4940-95-CL-B03 - List of Components

(Unless otherwise indicated, copies of the drawings and list of components are available from Armament, research, Development and Engineering Center, ATTN:SMCAR-EST-D,Rock Island, IL 61299-7300.)

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2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the document cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C 73.11 - Dimension of Plugs and Receptacles - General Purpose 125 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type
ANSI C 73.20 - Dimensions of Plugs and Receptacles - General Purpose 250 Volts, 15 Amperes, 2 Pole, 3 Wire, Grounding Type
ANSI C 73.61 - Dimension of Plugs and Receptacles - General Purpose 250 Volts, 50 Amperes, 3 Phase, 3 Pole, 4 Wire, Grounding Type

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B 117 - Spray Salt (Fog) Testing, Standard Method
ASTM D 3359 - Adhesion by Tape Test, Measuring

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG1 - Motor and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street N.W., Washington, D.C. 20037.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents are also available in or through libraries or other informational services.

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.)

3 REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish one complete shop for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification. Any changes or deviations by the contractor of the production shop from the approved first article during production shall be subject to the approval of the contracting officer.

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3.2 Design. The shop shall be in accordance with Top Assembly TA13217E1000 and as specified herein. The shop shall include all tools and equipment specified in the Set Components List as compiled from SC 4940-95-CL-B03 (see 6.3).

3.2.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the shop equipment. The contractor is responsible for preparing his own shop drawings.

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more shops for inspection as specified in 4.4.

3.4 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with best commercial practice, shall be suitable for the intended purpose in design and fabrication of the end item, and shall meet all the requirements herein. Materials shall be free from defects which would affect the performance, maintainability, reliability, durability, or longevity of the individual components or the end item of equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.5 Safety. All exposed parts, assemblies or accessories, which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are hazardous to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions.

3.6 Storage. The shop with the equipment shall be capable of withstanding an ambient temperature range of minus 50 degrees Fahrenheit (F) to plus 120 degrees F while in storage without encountering any adverse effects or damage (see 4.6.2.7.1).

3.7 Transportability.

3.7.1 Rail impact. The shop shall meet the rail impact test of MIL-STD-810 (see 4.6.2.5.2).

3.7.2 Lifting and tie-down. Lifting and tie-down provisions for fully loaded shop shall be in accordance with MIL-STD-209 (see 4.6.2.5.3).

3.7.3 Roadability. When loaded with the tools and equipment specified herein, the shop shall withstand driving on paved roads, unimproved roads, cross-country trails, a chatter course, and a diagonal ditch course. During or following driving, there shall be no damage to or displacement (see 6.8.3) of any component, accessory, part, or tool installed in or on the shop and no evidence of defective welded, bolted, or riveted connections on the shop.

3.7.4 Modes of transport.

3.7.4.1 Highway. The shop shall be within permit limits for highway transport in the United States.

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3.7.4.2 Rail. The shop shall meet the Association of Railroads (AAR) outline diagram.

3.7.4.3 Water. The shop shall be transportable on the LARC LX and larger vessels and ships.

3.7.4.4 Fixed wing aircraft.The shop shall be transportable in C-5 aircraft.

3.8 Truck chassis. The Government-furnished 5 ton, 6 x 6 truck chassis (see 3.19) shall be modified by the contractor in accordance with Top Assembly TA13217E1000 and as specified herein.

3.8.1 Shop weight. Unless otherwise specified (see 6.2), the gross vehicle weight (GVW) with all tools and equipment shall be no more than 36,000 pounds.

3.9 Body. The contractor shall fabricate and install the body on the chassis in accordance with the applicable drawings (see 3.2). The contractor shall furnish all items of equipment including the tools and equipment specified in 3.2.

3.9.1 Air compressor. An electric-motor-driven air compressor shall be furnished and installed by the contractor in accordance with the applicable drawings (see 3.2). The air compressor shall conform to MIL-C-13874, Type I, Class not applicable, Size 5 cubic feet per minute (cfm) at 175 pound per square inch gage (psig), less air receiver tank, fittings, and inflator device.

3.9.2 Electrical power system. The contractor shall furnish and install the entire electrical power system. The system shall be module configured consisting of an alternator, control panel and related controls. The modules shall be so designed as to provide accessibility for scheduled routine and minor unscheduled maintenance without complete removal. All lights, including pilot and panel lights, used during blackout conditions shall not emit energy between 660 and 1,100 nanometers of the electromagnetic spectrum.

3.9.2.1 Alternator. The alternator shall be of the brushless type conforming to NEMA MG1, insulation Class F. The alternator shall provide an output of 31.3 Kilovolt-ampere (KVA), 25 Kilowatt (KW), 120/208 Volt (V), 3 phase, 60 Hertz (Hz), on a 60 percent duty cycle with a regulation of 10 percent.

3.9.2.2 Control panel. The control panel shall be modular design and contain all controls and instruments (time totalizing meter, AC volt meter, percentage of load meter and frequency meters) necessary for the operation of the system. Controls shall be identified and functions indicated on the control panel. The control panel shall be accessible for circuit checks and component repair.

3.9.2.2.1 Frequency meter. The frequency meter shall be of the vibrating reed type with a range of 48-52 and 58-62 Hz. The meter shall have no less than 5 reeds at intervals not greater than 1 Hz. There shall be no dead zones between reeds.

3.9.2.2.2 Receptacles. Three 15 ampere, 125 volt, 2 pole, 3 wire receptacles, duplex, general purpose, conforming to the ANSI C 73.11 shall be furnished and installed in the control panel. The duplex receptacles shall be wired in pairs and four of the six receptacles protected with a ground fault interrupter (GFI) and located for access from ground level. Each pair of duplex receptacles shall be protected with a 15 ampere capacity circuit breaker. One 50 ampere, 250 volt, 3 pole, 4 wire receptacle, single, general purpose, 3 phase, conforming to ANSI C73.61 shall be furnished and installed in control panel and protected by a 50 ampere

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rcuit breaker. One 15 ampere, 250 volt, 2 pole, 3 wire, single phase, general purpose receptacle, conforming to ANSI C73.20 shall be furnished and installed in control panel and protected by a 10 ampere circuit breaker. The receptacles shall be identified to indicate the voltage supplied.

3.9.2.2.3 Circuit breakers. Three 250 volt, 3 pole, 3 wire, 3 phase circuit breakers, of 30 ampere capacity and one 15 ampere capacity shall be provided. The circuit breakers shall conform to W-C-375, single unit construction. Two of the 30 ampere circuit breakers shall be identified as "compressor" and "lathe". A means of preventing commercial power from being initiated while the power unit is on line shall be provided.

3.9.2.2.4 Time totalizing. A time totalizing digital meter in accordance with MIL-M-3971 shall be provided to register the number of hours of engine operation.

3.9.2.3 Grounding. A 1/4-20 threaded ground stud shall be provided on the lower front of the control panel as a terminal for ground leads from the control panel mounted equipment and the alternator.

3.9.2.4 Speed controls. The vehicle engine for the alternator shall be controlled by the governor to provide 60 Hz frequency within plus or minus 10 percent.

3.9.2.5 Overspeed. An electrical frequency from 63 to 66 HZ produced by the generator (alternator) shall be cause for engine shutdown.

3.10 Electrical system. Electrical systems shall be in accordance with the applicable drawings.

3.11 Electromagnetic interference (EMI). A permanent warning notice, indicating that EMI could result if electric power tools/equipment are operated in close vicinity to electronic communication or surveillance equipment, shall be included and displayed in a conspicuous place(s) on the shop van exterior adjacent to external power receptacle and on the master power control panel.

3.12 Tools and equipment. The tools and equipment shall be located as specified, anchored or stored to prevent movement (see 6.8.3) during transit. Tiedown devices shall be designed for a general tool or equipment configuration to allow interchange of like items from different manufacturing sources. Wherever possible, items shall be grouped together by function.

3.12.1 Fire extinguisher. The fire extinguisher shall be mounted inside the shop and shall be accessible from the center aisle and adjacent to the rear door.

3.13 Environmental conditions.

3.13.1 Precipitation. The shop shall withstand being exposed to natural or simulated rain falling toward the shop at a angle of approximately 45 degrees to the horizontal without water accumulating in the interior of the shop or damage to the components. The volume of rain impinging on the surface shall be not less that two inches in a five minute period. Simulated rain shall produce droplets of uniform distribution and velocity similar to natural rain at the point of contact with the shop.

3.13.2 Operating temperature. The shop shall perform as specified herein in any ambient temperature between -25 degrees Fahrenheit (F) and +120 degrees F.

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3.14 Sound level. The sound level of the shop shall be in accordance with MIL-STD-1474 Category D for steady-state noise when tested in accordance with 4.6.2.6. Hearing protectors need not be supplied with the shop equipment. Hazard signs, if required, shall state: "Caution, Hearing Protection Required" (see 6.12).

3.15 Fuels, lubricants, and coolants. The shop shall be assembled and tested with the products of the grade specified in Table I for the applicable ambient air temperature. The shop shall operate as specified herein when furnished with these products (see 6.6).

3.15.1 Oil lubrication. Unless otherwise specified herein (see 6.2), all components shall be filled to the operating level with military oils designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous place to indicate the temperature range and grade of lubricant used.

TABLE I. Service products.

Item	Application	Specification	Grade
1	Engine:		
	a. Heavy Duty	MIL-L-2104	10, 30
	b. Arctic	MIL-L-46167	One Grade
2	Fuel:		
	Diesel Fuel	VV-F-800	
3	Grease:		
	a. Automotive	MIL-G-10924	One Grade
	b. High Temperature	MIL-G-23827	One Grade
4	Hydraulic-system Oil:		
	a. Oil	MIL-L-2104	10
	b. Oil (low temperature)	MIL-L-46167	One Grade
5	Lubricating Oils:		
	Universal	MIL-L-2105	80, 90, 140
6	Antifreeze:		
	a. Ethylene glycol	O-A-548	One Grade
	b. Arctic	MIL-A-11755	One Grade
7	Brake Fluid	VV-B-680	One Grade

3.16 Fungus resistance. Electrical and electronic components and electrical insulation not otherwise specified herein shall be fungus inert, in accordance with requirement 4 of MIL-STD-454. Components of electrical circuitry that are not weather resistant to moisture and fungus growth shall be treated in accordance with MIL-T-152 and MIL-V-173.

3.17 Identification marking and instruction plates.

3.17.1 Identification marking. The unit shall be identified in accordance with MIL-STD-130.

3.17.2 Instruction plates. Instruction plates shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material.

3.18 Treatment and painting. The portions of the shop body and all other items requiring painting shall be cleaned, treated and painted with a chemical agent resistant coating (CARC) in accordance with finish 7.3.1 plus 20.24 or 24.17 of

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L-STD-171 for aluminum alloys or finish 5.1.1 plus 20.24 or 24.17 for ferrous alloys. Unless otherwise specified (see 6.2), color shall be 34094 of FED-STD-595. This treatment and painting requirement is a generic requirement. Treatment and painting requirements for component parts and minor assemblies may be otherwise specified. Application and testing of paint shall be in accordance with MIL-C-53072 and 4.6.2.12 herein. The external areas of the body shall be painted in a camamouflage pattern in accordance with drawing 13226E7428.

3.19 Government-furnished property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be furnished by the Government (see 6.5):

Item	Description	Identification	Qty/Shop
1	Chassis Truck, 5 ton, 6 x 6, 215 in. wheel base, variable governor, w/installed power take-off, and control assembly.	Model M944 Modified per CAGE 59678 Drawing 11023713	1 ea.
2	Heater, space, multifuel w/blower, 60,000 BTU/HR, mounting for discharge air-down position.	MIL-H-11511, Type 1 NSN 4520-00-683-8595	1 ea.
3	Padlock Set	MS21313-52 NSN 5340-00-682-1505	1 set
4	Engine lathe	MIL-L-23249, Type I, Size 1307	1 ea.

3.20 Reliability, availability and maintainability.

3.20.1 Reliability. The shop equipment shall demonstrate a mission reliability of at least 0.90.

3.20.2 Availability. The shop equipment availability shall be not less than 95 percent.

3.20.3 Maintainability. The shop equipment shall require a minimum of operational and in-storage maintenance. Replacement components will be used to the maximum extent possible. The shop equipment shall require a minimum of man-hours and skill to accomplish maintenance. Ease of maintenance and restricted use of components requiring special skills or tools for maintenance will be stressed. The maintenance ratio shall be no greater than 0.12 (see 6.8.1).

3.21 Workmanship.

3.21.1 Castings and forgings. Finished castings and forgings shall be free of fins, scale, inclusions, cold shuts, voids, cracks, thermal ruptures, laps, folds, mismatching, etc. No processes such as welding, peening, plugging, soldering, or pasting shall be used for repairing, reworking, or improving cosmetic appearance.

3.21.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of the material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of metal may be employed

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instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated surfaces shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges and corners are required or where they are not detrimental to safety.

3.21.3 Welding. All weldments shall conform to 11021065 drawing, type and class as specified.

3.21.4 Riveting. All riveted connections shall conform to 11021066 drawing, Class III, Table III.

3.21.5 Bolted connections. Bolt holes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching threaded area securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, tested, and inspected in accordance with referenced specifications and standards.

4.1.2 Disassembly inspection. The first article shop failing a performance test shall be disassembled in the presence of a Government representative to the extent necessary to determine the cause of the failure. Each disassembled part shall be inspected in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, and tolerances. Parts not complying with such requirements shall be rejected.

4.1.3 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Initial production inspection (see 4.4).
c. Quality conformance inspection (see 4.5).
d. Inspection comparison (see 4.7).
e. Packaging inspection (see 4.8).

4.3 First article inspection (see 3.1).

4.3.1 Examination. The shop shall be examined for the defects marked 'X' in Column 1 of Table II. Presence of one or more defects shall be cause for rejection.

TABLE II. Examination schedule.

First Article	Initial Production	Quality Conformance	Defect	Requirement Paragraph
X	X	X	Assembly incomplete or incorrect.	3.2, 3.2.1
X	X	-	Coated cloth inclosures do not fit as specified when erected.	3.2
X	-	X	Material not as specified.	3.4
X	X	X	Safety devices and guards missing or not as specified.	3.5
X	-	X	Dimensions and weight not as specified.	3.7.4.1 thru 3.7.4.4, 3.8, 3.8.1
X	X	X	Body and equipment not as specified.	3.9 thru 3.10, 3.19
X	X	X	Parts, components, tools or equipment missing, or not functioning as specified.	3.12, 3.12.1
X	-	X	Stowage and tiedown not as specified.	3.12
X	X	X	Lubrication not as specified.	3.15, 3.15.1
X	-	X	Fungus and moisture resistance not as specified.	3.16
X	X	X	Identification, marking, warning and instruction notices and plates not as specified.	3.11, 3.17
X	-	X	Treatment and painting not as specified.	3.18
X	X	X	Workmanship not as specified.	3.21

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4.3.2 Tests. The shop shall be tested as specified in Column 1 of Table III. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

TABLE III. Test schedule.

First Article	Initial Production	Quality Conformance	Test	Test Paragraph	Requirement Paragraph
1	2	3	4	5	6
X	X	X	Heating system	4.6.2.2	3.19
X	-	-	Precipitation		3.13.1
X	X	X	Maximum power operation	4.6.2.4.1	3.9.2.1
X	X	X	Installed equipment and power tools	4.6.2.4.2	3.9.2, 3.12
X	X	X	Overspeed	4.6.2.4.3	3.9.2.5
-	X	-	Road tests	4.6.2.5.1	3.7.3
X	-	-	Rail impact	4.6.2.5.2	3.7.1
X	-	-	Slinging provisions	4.6.2.5.3.1	3.7.2
X	-	-	Tie-down provisions	4.6.2.5.3.2	3.7.2
X	-	-	Sound level	4.6.2.6	3.14
X	-	-	Environmental	4.6.2.7.1	3.6, 3.13.2
-	X	-	Reliability	4.6.2.8	3.20.1
-	X	-	Availability	4.6.2.10	3.20.2
-	X	-	Maintainability	4.6.2.11	3.20.3
X	-	X	Treatment and Painting	4.6.2.12	3.18

4.4 Initial production inspection. When specified (see 3.3), one or more initial production shops will be selected at random by the Government from shops being produced from production tooling and will be examined for the defects marked "X" in Column 2 of Table II and subjected to the tests marked "X" in Column 2 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production shop shall not exclude the remaining shops from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any and all other tests contained in this specification as part of the initial production test and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests. Prior to examination or test, the initial production shop shall be lubricated with oils and greases specified in 3.15. Oils shall be those designated for use in the ambient temperature at which the test is being performed.

4.4.1 Inspection failure. Failure of an initial production shop to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on shops previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all shops accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to contracting officer.

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4.5 Quality conformance inspection.

4.5.1 Examination. Each shop shall be examined for the defects marked "X" in Column 3 of Table II. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each shop shall be subjected to the tests marked "X" in Column 3 of Table III. Failure of any test shall be cause for rejection.

4.6 Inspection schedule.

4.6.1 Examination. Examination shall be in accordance with Table II.

4.6.2 Tests.

4.6.2.1 Test schedule. Tests shall be in accordance with Table III. Sequence of the test at the option of the test facility except as required in test paragraphs.

4.6.2.2 Heating system test. Operate the heater continuously for not less than 15 minutes to check the operation of the heater, thermostat, and heat deflectors. Any of the following that occurs as a result of faulty material supplied by the supplier or incorrect installation by the supplier shall constitute failure of this test:

a. Malfunctioning of deflector or fuel supply line.
b. Smoking, sooting, or back draft.
c. Entrance of fuel vapors or exhaust gases into the interior of the shop.

4.6.2.3 Precipitation test. Park the shop on a horizontal surface and expose the shop to natural or simulated rain falling toward the shop at an angle of approximately 45 degrees to the horizontal. The volume of rain impinging on the surface shall be not less than 2 inches in a 5 minute period. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the shop. Conduct the test as follows:

a. Expose the top and left side to the rain as described above for a period of 5 minutes.
b. Repeat Step "a" with the top and front exposed.
c. Repeat Step "a" with the top and right side exposed.
d. Repeat Step "a" with the top and rear exposed.

Water accumulation in the interior of the shop or damage to components shall constitute failure of this test. The canvas enclosure shall not be tested.

4.6.2.4 Electrical power system.

4.6.2.4.1 Maximum power operation. Load the unit to maximum rated load of 208-volt, 3 phase, 4 wire, 25 kw, AC power. Apply a straight resistance AC load for 2 hours in 10 minute cycles of continuous load for 6 minutes followed by 4 minutes at no load. Nonconformance of the electrical components to 3.9.2 or failure of the belts or other mechanical components shall constitute failure of this test. The quality conformance test shall consist of continuous load for 6 minutes and may be performed as a part of 4.6.2.4.2.

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4.6.2.4.2 Installed equipment and power tools. Using both external and shop generated power, operate all installed equipment and power tools. Failure of the power unit to compensate for high starting currents or inability of installed equipment or power tools to function shall constitute failure of this test.

4.6.2.4.3 Overspeed. Conduct this test at the conclusion of the test specified in 4.6.2.4.2. Pull the governor control to increase engine speed until the overspeed control stops the engine. Observe the AC frequency at which the overspeed control shuts-down the engine. Failure of the overspeed control to stop the engine when the frequency exceeds 66 Hz or inability to restart the engine following resetting of the control shall constitute failure of this test.

4.6.2.5 Transportability.

4.6.2.5.1 Road tests. Load the shop with the tools and equipment specified herein and perform the following tests. Both prior to and after the road test, perform the installed equipment and power tool test specified in 4.6.2.4.2.

a. Unimproved roads. Drive the shop 350 miles on unimproved roads at maximum safe speeds. The course shall consist of poorly maintained gravel and dirt roads with moderate washboarding and rutting and occasional potholes.
b. Cross-country trails. Drive the shop 150 miles on cross-country trails at maximum safe speeds. The course shall consist of native soil with rock added as required to maintain passability. Surface conditions shall range from smooth to rough with washboarding, rutting, potholes and depressed areas.
c. Chatter course. Drive the shop over a chatter course until at least 2,600 either upward or downward accelerations of magnitudes specified herein have been applied to a rear wheel center. The course shall be of construction that the acceleration magnitudes indicated below can be obtained by adjusting the vehicle speed without the frequency of the accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude indicated below shall not be exceeded, the accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives in one pass over the course at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specified herein. Measure the accelerations by a trial test to establish the required vehicle test speed. Then disconnect the recording instruments and complete the test at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually an axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the acceleration shall be not less than 4 times the acceleration of gravity (G) nor more than 7G.
d. Diagonal ditch course. Drive the shop over a series of ditches until 600 reversal torsional frame stresses have been applied. The ditch profile shall be such as to allow vertical wheel travel of 18 inches from crest to root. The ditches shall be at 45 degrees to the centerline of the course and spaced a sufficient distance apart to allow only one rear wheel and one front wheel of opposite sides in the ditches at any one time.

Examine the shop at the completion of each phase of the road test. Damage to or displacement (see 6.8.3) of any component, accessory, part or tool; evidence of defective welded, bolted, or riveted connections, or accumulation of dust, mud or water in the interior, shall constitute failure of these road tests.

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4.6.2.5.2 Rail impact. The shop shall be subjected to the rail impact test of MIL-STD-810 (see 3.7.1). The rail impact procedure shall be in accordance with MIL-STD-810 (located underneath "Shock" with the following additions and exceptions:

a. Using shop power, operate all power tools for at least five (5) minutes. Operate the air compressor for at least 15 minutes. Discharge air from the reservoir and allow the pressure to build back up to at least 100 psi several times during the test. Operate the alternator for at least 15 minutes during which time test the alternator to determine if it will deliver maximum power. Test the welder by running a six (6) inch bead at near maximum capacity and a six (6) inch bead at near minimum capacity.
b. The shop shall be positioned lengthwise on a railroad car. A dummy load shall not be used. The tools and equipment shall be located as specified herein, anchored or stored to prevent movement during transit. Examine the shop for any physical damage after each impact and record the results.
c. After the impact tests are completed repeat tests in 4.6.2.5.2,a.
d. Any evidence of damage or failure of any component to perform as specified shall be cause for rejection.

4.6.2.5.3 Lifting and tie-down test. To determine compliance with 3.7.2 and MIL-STD-209, the fully loaded shop shall be subjected to the following tests. The loads are based on a maximum shop weight of 36,000 pounds (see 3.8.1).

4.6.2.5.3.1 Slinging provisions. A static pull to no less than the required design load shall be conducted on each lifting provision for no less than 90 seconds in a direction of pull 45 degrees upward from the shop floor toward the apex where the four cables meet in flight. All provisions do not have to be tested at the same time. Tests shall be performed in accordance with MIL-STD-209. The design loads for the four lifting provisions shall be as follows:

a. Front two provisions - subject each provision to a 46,387 pound static pull.
b. Rear two provisions - subject each provision to a 19,062 pound static pull.

Any permanent deformation, set, or structural damage in the provisions or other equipment structural components resulting from application of the loads to the provisions shall be cause for rejection.

4.6.2.5.3.2 Tie-down provisions. A static pull to no less than the required design load shall be conducted on each tie-down provision for no less than six (6) seconds in the direction of pull indicated. All provisions do not have to be tested at the same time. Tests shall be performed in accordance with MIL-STD-209. The design loads and direction of force for the eight tie-down provisions shall be as follows:

a. Rear six provisions - subject each provision to a 24,000 pound static pull in a rear longitudinal direction (horizontal toward the rear, parallel to edge of bed).
b. Front six provisions - subject each provision to a 24,000 pound static pull in a forward longitudinal direction.
c. All eight provisions - subject each provision to a 9,000 pound static pull in a vertically downward direction.
d. All eight provisions - subject each provision to 13,500 pound static pull in a lateral direction (horizontally outward, perpendicular to longitudinal force in a.)

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Any permanent deformation, set, or structural damage in the provisions or other equipment structural components resulting from application of the loads to the provisions shall be cause for rejection.

4.6.2.6 Sound level measurement. Operate the shop at governed speed with the alternator and air compressor operating. Measure the sound level at the tailgate work station while operating each of the power tools. Sound level in excess of Category D of MIL-STD-1474 shall require the installation of hazard signs as specified in 3.14.

4.6.2.7 Environmental.

4.6.2.7.1 Temperature extremes operation and storage tests. Place the shop in an ambient air temperature at minus 50 degrees plus or minus 5 degrees (F) for a minimum period of 72 hours. At the end of this time period raise the temperature to minus 25 degrees F plus or minus 5 degrees for a period of 12 hours. At the end of this time period the following operational tests shall be performed (see 3.13.2).

a. Inspect the entire unit for any damage which may have resulted from exposure to temperature extreme.
b. Start and operate the alternator. It shall provide electrical power to the 120/240 volt system.
c. Start and operate the heating system using diesel fuel Class DF-A having a cloud point not lower than minus 55 degrees F (see MIL-H-11511). The heater shall operate normally.
d. Perform 4.6.2.4.2 (Installed equipment and power tool). All equipment shall start and operate normally.

After the conclusion at the foregoing test, stop all equipment, close up the unit and raise the ambient temperature to 120 degrees F plus or minus 5 degrees F for a minimum period of 72 hours. Repeat step a, b, c, and d. Failure of any component, system or tool shall be cause for rejection of the unit.

4.6.2.8 Reliability. The initial production shop shall be tested for a total of 10 missions as described in 4.6.2.9, to determine conformance to 3.20.1. Data shall be gathered to evaluated the system reliability. Reliability is to be evaluated from MTBF and mission operating time (see 6.8.2 and 6.8.8).

$$\text{Reliability} = e^{-t/u}$$

where,

t = mission operating time, hours
u = mean time between failures (MTBF), hours
e = 2.718

The shop shall be considered to have demonstrated the required reliability if zero chargeable reliability failures occur (see 3.20.1 and 6.8.9).

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4.6.2.9 Mission operation and test. A mission shall consist of the following:

a. Drive the fully loaded shop set at maximum safe speeds for ten miles on unimproved roads and ten miles on cross-country trails.
b. Using the instructions furnished with the shop, set-up the shop for operation to include rigging out of tent enclosure on the initial and final missions.
c. Operate each installed equipment and power tool for a period not less than ten minutes one time per mission, except as noted.
d. Prepare the shop for movement to another location, to begin the next mission.

The following list of tools, which are included in the shop, shall be subjected to the 10 minutes test per mission:

a. Air compressor.
b. Charger, battery.
c. Drill, electric, portable, 3/8 in.
d. Drill, electric, portable, 1/2 in.
e. Drilling machine.
f. Floodlight, electric.
g. Grinding machine, machine tool attachment.
h. Grinding machine, utility.
i. Lathe, engine.
j. Light, extension, 3-conductor.
k. Light, power tool.
l. Milling and drilling head.
m. Moisture stabilizer, welding electrode.
n. Power, system, electrical.
o. Sander, disk, electric portable.
p. Soldering iron, electric.
q. Wrench, impact, electric.
r. AC welder and accessories

4.6.2.10 Availability. During performance of the reliability test (see 4.6.2.8) data shall be gathered to evaluate the systems availability. Availability shall be evaluated with respect to operating time and corrective maintenance (see 6.9.9 and 6.9.8).

$$\text{Availability} = \frac{(\ \text{MTBUMA}\ )}{(\text{MTBUMA}+\text{MTTR})} \quad (100 \text{ percent})$$

Where,

MTBUMA = Mean time between unscheduled maintenance actions (unscheduled means corrective maintenance actions)

MTTR = Mean time to repair (use only times associated with corrective maintenance) (see 6.8.6)

The system availability shall be 95 percent (see 3.20.2). Nonconformance with 3.20.2 shall constitute failure to meet the availability requirement.

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4.6.2.11 Maintainability. Maintenance operations required during first article and initial production tests shall be gathered to evaluate the system maintainability. Maintainability shall be evaluated with respect to total maintenance man-hours and total operating time during testing.

$$\text{Maintenance Ratio} = \frac{TNMMT}{TOT}$$

where,

TNMMT = Total number of man-hours of maintenance (scheduled maintenance and unscheduled repairs) during testing.

TOT = Total operating time (see 6.8.8).

Any support elements such as technical publications, repair parts, special tools, etc., provided under the contract for assisting or conducting maintenance shall be evaluated. The system maintenance ratio shall be no greater than 0.12 (see 3.20.3). Nonconformance with 3.20.3 shall constitute failure of this test.

4.6.2.12 Treatment and painting.

4.6.2.12.1 Ferrous material. Application of finish 5.1.1 plus 20.24 or 24.17 of MIL-STD-171 (see 3.18). The following test methods and procedures shall be applied to parts on the Shop Equipment, Organizational Repair, Truck-Mounted, composed of ferrous material.

4.6.2.12.1.1 Cleaning. Cleaning shall be performed in accordance with test methods specified in TT-C-490 for removal of soils and corrosion products; rinsing; and final rinse. The total acid and alkali levels shall not exceed those of the rinsing test method of TT-C-490. The cleaning test shall be conducted daily on a minimum of (4) four production items, representative of the production process.

4.6.2.12.1.2 Phosphate coating weight. The phosphate coating weight shall be verified using the phosphate controls test of TT-C-490. The phosphate controls test shall be met before the water break test is performed on parts. The phosphate coating weight test shall be conducted on three specimens after a maximum of four (4) hours of phosphate processing. Test specimens shall be prepared accordance with TT-C-490.

4.6.2.12.1.3 Water break. A water break test shall be conducted in accordance with the condition of surface test of MIL-C-53072. All condition of service test requirements shall be met prior to painting. The condition of surface test shall be conducted on all parts, (100 percent) before primer is applied.

4.6.2.12.1.4 Primer dry film thickness. The primer dry film thickness shall be verified using the primer dry film thickness test of MIL-C-53072. The paint thickness shall conform to the minimum values given in the dry film thickness table in MIL-C-53072. Dry film thickness for MIL-D-23377 primer shall be 0.6 to 0.9 mils. This test shall be conducted daily on at least four (4) production items representative of the production process.

4.6.2.12.1.5 Primer salt spray. A primer salt spray test shall be performed in accordance with ASTM B 117 using a 5 percent solution for 336 hours. There shall be no rusting visible to the unaided eye. The items to be tested shall be representative of the item processed (for example, distributed randomly and

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ocessed concurrently with the production items). For a batch type paint process, four (4) items shall be tested from each week (40 hours) of production or portion thereof. Scrap items or portions thereof may be used for this test. (Large items may be cut to a size suitable for the salt spray chamber.)

4.6.2.12.1.6 Primer and topcoat adhesion. A primer and topcoat adhesion test shall be performed in accordance with the adhesion test of MIL-C-53072 and ASTM D 3359, Method B, cross cut tape adhesion, using the six line pattern and 2 millimeter spacing. Acceptance/rejection criteria shall be as specified under the adhesion test requirements of MIL-C-53072. The adhesion test shall be performed on the primer coat and also after the topcoat is applied. The adhesion test shall be conducted on one item per batch, or if a continuous process is used, one item per week (40 hours, or any portion thereof).

4.6.2.12.1.7 Topcoat dry film thickness. Topcoat dry film thickness shall be verified using the dry film thickness test of MIL-C-53072. Use the same test specimen as used for the primer dry film thickness test (see 4.6.2.12.1.4). Measurements shall be taken in the same locations as where the primer dry film thickness test measurements were taken. The difference between the total coating thickness measurement and the primer thickness measurement is the thickness of the topcoat. Topcoat film thickness shall conform to the requirements in MIL-C-53072 for paint used. Topcoat dry film thickness test shall be conducted daily or at leas four (4) production items representative of the production process.

4.6.2.12.1.8 Topcoat solvent wipe. A topcoat solvent wipe test in accordance with MIL-C-53072 shall be performed and all requirements therein shall be met prior to acceptance. The topcoat solvent wipe test shall be performed on four (4) items selected from each batch, or, for a continuous process, four (4) items shall be selected from each week (40 hours of production, or any portion thereof).

4.6.2.12.2 Aluminum material. Application of finish 7.3.1 plus 20.24 or 24.17 of MIL-STD-171 (see 3.18). The following test methods and procedures shall be applied to parts on the Shop Equipment, Organizational Repair, Truck-Mounted, composed of aluminum material.

4.6.2.12.2.1 Cleaning. Cleaning shall be performed in accordance with test methods specified in TT-C-490 for removal of soils and corrosion products; rinsing; and final rinse. The total acid and alkali levels shall not exceed those of the rinsing test method of TT-C-490. The cleaning test shall be conducted daily on a minimum of (4) four production items, representative of the production process.

4.6.2.12.2.2 Corrosion test - pretreatment. The corrosion test - pretreatment shall be performed after pretreatment is applied in accordance with the corrosion resistance test in MIL-C-5541D. Corrosion test - pretreatment requirements shall be met before the water break test is performed on the parts. The items to be tested shall be representative of the item processed (for example, distributed randomly and processed concurrently with the production items). For a batch type paint process, four (4) items shall be tested from each week (40 hours) of production or portion thereof. Scrap items or portions thereof may be used for this test. (Large items may be cut to a size suitable for the salt spray chamber.)

4.6.2.12.2.3 Water break. A water break test shall be conducted in accordance with the condition of surface test of MIL-C-53072. All condition of service test requirements shall be met prior to painting. The condition of surface test shall be conducted on all parts, (100 percent) before primer is applied.

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4.6.2.12.2.4 Primer dry film thickness. The primer dry film thickness shall be verified using the primer dry film thickness test of MIL-C-53072. The paint thickness shall conform to the minimum values given in the dry film thickness table in MIL-C-53072. Dry film thickness for MIL-D-23377 primer shall be 0.6 to 0.9 mils. This test shall be conducted daily on at least four (4) production items representative of the production process.

4.6.2.12.2.5 Primer adhesion. Primer adhesion shall be tested in accordance with the adhesion (wet) tape test of MIL-C-5541D. No intercoat seperation shall occur between the paint system and the conversion coating and the base metal. Any separation shall be cause for rejection. The primer adhesion test shall be conducted on one item per batch, or if a continuous process is used, one item per week (40 hours), or portion thereof.

4.6.2.12.2.6 Primer salt spray. A primer salt spray test shall be performed in accordance with ASTM B 117 using a 5 percent solution for 336 hours. There shall be no rusting visible to the unaided eye. The items to be tested shall be representative of the item processed (for example, distributed randomly and processed concurrently with the production items). For a batch type paint process, four (4) items shall be tested from each week (40 hours) of production or portion thereof. Scrap items or portions thereof may be used for this test. (Large items may be cut to a size suitable for the salt spray chamber.)

4.6.2.12.2.7 Primer and topcoat adhesion. A primer and topcoat adhesion test shall be performed in accordance with the adhesion test of MIL-C-53072 and ASTM D 3359, Method B, cross cut tape adhesion, using the six line pattern and 2 millimeter spacing. Acceptance/rejection criteria shall be as specified under the adhesion test requirements of MIL-C-53072. The adhesion test shall be performed on the primer coat and also after the topcoat is applied. The adhesion test shall be conducted on one item per batch, or if a continuous process is used, one item per week (40 hours, or any portion thereof).

4.6.2.12.2.8 Topcoat dry film thickness. Topcoat dry film thickness shall be verified using the dry film thickness test of MIL-C-53072. Use the same test specimens as used for the primer dry film thickness test (see 4.6.2.12.1.4). Measurements shall be taken in the same locations as where the primer dry film thickness test measurements were taken. The difference between the total coating thickness measurement and the primer thickness measurement is the thickness of the topcoat. Topcoat film thickness shall conform to the requirements in MIL-C-53072 for paint used. Topcoat dry film thickness test shall be conducted daily or at leas four (4) production items representative of the production process.

4.6.2.12.2.9 Topcoat solvent wipe. A topcoat solvent wipe test in accordance with MIL-C-53072 shall be performed and all requirements therein shall be met prior to acceptance. The topcoat solvent wipe test shall be performed on four (4) items selected from each batch, or, for a continuous process, four (4) items shall be selected from each week (40 hours of production, or any portion thereof).

4.6.2.12.3 Ferrous and aluminum material. Application of finish 5.1.1 or 7.3.1 plus 20.24 of MIL-STD-171 (see 3.18). The following test methods and procedures shall be applied to assemblies on the Shop Equipment, Organizational Repair, Truck Mounted consisting of mixed parts of both ferrous and aluminum materials. The ferrous parts that are installed into these assemblies shall be finished in accordance with 4.6.2.12.1 before assembly.

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4.6.2.12.4 Certificate of performance - paint. Primer paint lots furnished with each specification shall be provided with a Certificate of Performance verifying that the lot meets the requirements specified herein. Topcoat paint lots shall be furnished with a certification from the Belvoir Research, Development and Engineering Center, Ft. Belvoir, VA, verifying that the material meets the near infrared reflectance properties per MIL-C-53072. Manufactured date and lot number shall be included in the information in the certification. Topcoat paint lots furnished with each specification shall be provided with a Certificate of Conformance verifying that the lot meets the requirements herein together with the near infrared reflectance certification.

4.7 Inspection comparison. The Government may select shops at any time during the contract production period and subject these shops to the examination specified in Column 2 of Table II and the tests specified in Column 2 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Shops will be selected at random from those which have been accepted by the Government and will not include the previously inspected first article or initial production model. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison shop to meet any requirement specified herein during and as a result of the examination and test pecified in 4.7 shall be cause for rejection of the inspection comparison shop and shall be cause for refusal by the Government to continue acceptance of production shops until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier on shops previously produced and accepted under the contract. Any deficiencies found as a result of inspection comparison inspection will be considered prima facie evidence that all shops accepted after approval of the initial production shop are similarly deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.8 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the requirements of Section 5 herein.

5 PACKAGING

5.1 Preservation. Preservation shall be Level A, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservatives and methods. Preservatives (P) and methods of preservation (1C-1) specified herein shall conform to and be applied in accordance with the requirements of MIL-P-116.

5.1.1.2 Sets or kits. When items are furnished in sets or kits with a carrying case, the items shall be preserved as specified herein and, when possible, replaced in the case.

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5.1.1.3 Items having mounting brackets or holders. Items furnished with mounting brackets or holders shall be preserved as specified herein and, when possible, replaced in the bracket or holder and secured to prevent movement and damage.

5.1.1.4 Unprotected surfaces. Unprotected exterior metal surfaces of the components, not specifically specified herein, requiring the application of a contact preservative in accordance with MIL-P-116 shall be preserved as follows.

5.1.1.4.1 Unfinished surfaces. Unfinished surfaces shall be coated with Type P-1 preservative except that cold finished and hot rolled steel shall be coated with Type P-7 preservative.

5.1.1.4.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with Type P-2, P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2 and secured in place with tape conforming to PPP-T-60, Type IV.

5.1.1.5 Truck chassis. The truck chassis with body mounted shall be preserved in accordance with the preservation and packaging requirements of MIL-V-62038 for a mobile type shipment with the following exceptions and additional requirements.

5.1.1.5.1 Engine. The engine and engine accessories shall be preserved in accordance with the Level A requirements of MIL-E-10062, Type I, Method I.

5.1.1.5.2 Batteries, cables and electrolyte. Dry charged batteries shall be installed and secured in appropriate battery carrier. Filler cap openings shall be closed. Battery cables shall be secured to the carrier with 3/4 inch tape conforming to type IV of PPP-T-97. The electrolyte shall be packaged with the Level A requirements of O-S-801.

5.1.1.5.3 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-V-62038.

5.1.1.6 Shop enclosure. Each section of the coated cloth shall be compactly and neatly folded, then wrapped with paper conforming to PPP-B-1055, Class C1 and the wrap secured with tape conforming to PPP-T-60, Type IV. The tent pins shall be packaged in a close-fitting box conforming to PPP-B-636. The coated cloth and tent pins shall be placed in the storage compartment provided. The unassembled framing shall be placed and secured on the roof as provided for.

5.1.1.7 Alternator. The alternator shall be preserved in accordance with MIL-E-16298, Level A, using the alternate method as specified for assembled machines.

5.1.1.8 Compressor. The air compressor and components shall be preserved and packaged in accordance with MIL-C-3600, Level A.

5.1.1.9 Hand tools and tool sets. Hand tools and tool sets shall be preserved in accordance with the Level A requirements of PPP-P-40.

5.1.1.10 Belts and pulleys. All drive belts shall be released from tension or removed. Unpainted, unplated ferrous metal surfaces of the pulley shall be coated with primer conforming to TT-P-664. Removed belts shall be secured convenient to the pulley for which intended.

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5.1.1.11 Space heater. The heater shall be completely drained of fuel.

5.1.1.12 Machine tools, attachments, and accessories. Machine tools such as the lathe, milling attachment, hydraulic press, bench grinder, grinding machine, drilling machine and other attachments, components, and accessories shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-M-18058. MIL-M-18058 shall also be used as a guide in the selection of preservatives and materials to be applied to items of similar construction not listed herein.

5.1.1.13 Chain hoist. The chain and any other unpainted metal surface of the hoist shall be coated with Type P-2 preservative. The hoist shall be wrapped in paper conforming to MIL-B-121, Grade A, Type I, Class 2, and packaged in a close-fitting box conforming to PPP-B-636, Domestic Type.

5.1.1.14 Items furnished in commercial containers. Items such as pigment blue, sealing compound, welding electrodes and any other items commercially furnished in sealed cans, pails, drums, jars, or tubes shall not require preservation.

5.1.1.15 Method III. The fitting kit, rubber hose, insert and assembly kit, electrical connector repair kit, temperature indicating compound, hardware items, and coupling halves shall be individually packaged Method III.

5.1.1.16 Method IC-1. Technical publications and connector adapters shall be preserved Method IC-1.

5.1.1.17 Method IC-2. The first aid kit, flood light, hydraulic pump, spindle adapter, valve guide brush, and spark plug cleaner shall be individually preserved Method IC-2. The unit container shall be the contractor's commercially designed container or a close-fitting box conforming to PPP-B-566 or PPP-B-636, Class Domestic-variety, grade, and style optional.

5.1.1.18 Electrical wire. Electrical wire shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-C-12000.

5.1.1.19 Cable assemblies and electrical leads. Each cable assembly or test lead shall be coiled to a minimum safe diameter, with no kinks, bends or twists and secured with cotton twine or rope.

5.1.1.20 Hydraulic jacks and lifts. The hydraulic-fluid supply tanks of hydraulic control systems shall be filled to the operating level with hydraulic fluid required by the technical manual supplied with the equipment. The systems shall be operated to insure complete coverage of the systems with the fluid. Unpainted exterior surfaces shall be coated with Type P-1 preservative.

5.1.1.21 Locks and keys. The interior of locks shall be coated with dry lubricant conforming to MIL-M-7866. The locks locks shall be operated to assure penetration of the lubricant. The locks used for securing the doors shall be protected against the elements by wrapping the locks in barrier material conforming to MIL-B-121, Grade A. The wrap shall be secured with tape conforming to PPP-T-60, Type IV. The keys shall be preserved Method III and forwarded to destination separately by registered mail. A cloth or paper shipping tag containing the shop equipment identification number and the trailer registration number shall be attached to the related padlock keys for future positive identification.

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5.1.1.22 Fire extinguisher. Unpainted exterior metal surfaces of the fire extinguisher requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. The extinguisher shall be placed in the mounting bracket and secured.

5.1.1.23 Electronic and electrical equipment. Electronic and electrical equipment and accessories, shall be preserved and packaged in accordance with MIL-E-17555, Level A.

5.1.1.24 Cork and paper gasket sheets. Sheets of like description and size shall be rolled together to the minimum safe diameter and each roll wrapped in barrier material conforming to PPP-B-1055, class C-1.

5.1.1.25 Welding and soldering equipment. Welding and soldering equipment shall be preserved in accordance with the Level A preservation and packaging requirements of MIL-W-45562.

5.1.1.26 Other tools and equipment. Tool and equipment items which are not specifically covered by the packaging requirements of this specification shall be preserved and unit packaged in accordance with the applicable commodity specification of MIL-T-45542.

5.1.1.27 Miscellaneous supplies and equipment. Items, equipment, and components not specifically identified herein, shall be preserved as specified for supplies and equipment of similar design and construction.

5.1.1.28 Consolidation. Components that cannot be placed in the drawer, toolboxes, or compartments, or secured by the mounting brackets or holders provided, shall be consolidated together in close-fitting boxes conforming to PPP-B-636 Domestic Type, variety, grade and style optional. The boxes shall be of a size to fit into the available space within the body. The contents shall be blocked, braced, and cushioned as applicable to prevent movement and damage to the contents within the box.

5.1.2 Level C. All items and equipment shall be preserved and packaged in accordance with MIL-STD-1190.

5.2 Packing. The truck mounted shop, with equipment enclosed, shall be prepared for shipment mobile in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings. Items and equipment, installed in the shop body, and any other movable equipment or items, shall be cushioned, blocked, braced, and anchored to prevent damage to the equipment or to the body. All doors and drawers shall be secured to prevent accidental opening and in a manner to prevent pilferage of contents. Padlock keys shall be identified with the vehicle registration number and shall be sent by registered mail to the recipient of the shop equipment (see 6.10).

5.3 Marking. Marking for all levels shall be in accordance with MIL-STD-129. Bar code marking is also required in accordance with MIL-STD-129.

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6 NOTES

(This section contains information of a general explanatory nature that may be helpfull, but is not mandatory.)

6.1 Intended use. The organizational repair shop is intended for use by personnel engaged in the maintenance of military mechanical equipment.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of the first article (see 3.1).
c. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.1).
d. When initial production inspection is required and number of shops to be furnished (see 3.3).
e. Gross weight if other than that listed in 3.8.1.
f. Components not filled with oil (see 3.15.1).
g. Paint color, if different (see 3.18).
h. Government-furnished property if other than those listed in 3.19.
i. Level of preservation and packaging required (see 5.1).

6.3 The contracting officer should arrange to furnish information concerning items specified in set SC 4940-95-CL-B03 (see 3.2).

6.4 First article. Any changes or deviations of production shops from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the supplier of his obligation to furnish shops conforming to this specification.

6.5 Technical interface data. The contracting officer should arrange to provide information concerning technical interface for the government furnished truck chassis specified in 3.19.

6.6 Lubricants. When requested by the contractor, the contracting officer should obtain a list of military lubricants applicable to the shop covered by this specification (see 3.15).

6.7 Government-furnished property. The contracting officer should arrange to furnish the supplier the property listed in 3.19.

6.8 Definitions.

6.8.1 Maintenance ratio. The ratio of the number of maintenance man-hours required for both scheduled maintenance and unscheduled repairs, to the number of hours of operation.

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6.8.2 Mean-time-between-failure (MTBF). For a particular interval, the total functioning life of a population of an item divided by the total number of failures within the population during the measurement interval. The definition holds for time, cycle, miles, events, or other measure of life units.

6.8.3 Displacement. Displacement is defined as dislodgement of any tool or item of equipment from its designated position. Movement of a tool or item of equipment within its designated position without any dislodgement is permitted (see 3.7, 3.12 and 4.6.2.5).

6.8.4 Availability (achieved). The total operating time divided by the total operating time plus number of maintenance clock hours required for both scheduled maintenance and unscheduled repairs.

6.8.5 Availability (inherent). The ratio of MTBF to MTBF plus MTTR.

6.8.6 Mean time to repair (MTTR). The total corrective maintenance time in clock hours divided by the total number of corrective maintenance actions.

6.8.7 Corrective maintenance action. Any test incident that requires unscheduled maintenance time and includes both reliability chargeable failures and nonchargeable incidents.

6.8.8 Total operating time (TOT). Defined to be an 8-hour period per mission. This time does not include scheduled maintenance or unscheduled repair time. If the mission profile is completed in less than 8 hours, the balance is considered to be standby time.

6.8.9 Chargeable reliability failure. A chargeable reliability failure is defined as any malfunction:

a. Which cannot be corrected by organizational repair within five minutes by adjustment, repair, or replacement action using controls on the equipment, tools, or parts.

b. Which causes or may cause:

(1) Failure to start operation.
(2) Cessation of operation.
(3) Degradation of performance below specified levels.
(4) Damage to the shop by continued operation.
(5) Safety hazards to personnel.

6.9 Packing. Levels of packing have not been included for the truck mounted shop equipment as the nature of the item is such that only mobile packing is appropriate. For purposes of preservation and packing level markings, the level of packing shall be considered Level A and shall be marked A/A, B/A, or Industrial, as applicable (see 5.2).

6.10 Contract data requirements. Required technical data such as operator's manuals, parts lists, schematics, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

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6.11 Provisions. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of shops.

6.12 Ear protection. When the noise level exceeds Category D of MIL-STD-1474, the operator's manual for the shop shall include appropriate instructions for use of hearing protection when using equipment that produces noise above 85db(A) inside and outside of the shop, and within a distance from the shop at which 85db(A) will be met.

6.13 Subject term (key word) listing.

Material
Safety
Shop equipment
Transportability

Custodian:
Army - AL

Preparing activity:
Army - AL

Project No. 4940-A602

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

PN 4940-A602
MIL-S-45537, Rev F-Shop Equipment, Organizational Repair, Truck Mounted

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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